Qwahn D. Kent
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Education

Princeton University, Department of Ecology and Evolutionary Biology, Princeton, NJ
Doctor of Philosophy in Ecology and Evolutionary Biology

Cornell University, College of Agriculture and Life Sciences, Ithaca, NY
Bachelor of Science in Biological Sciences (Honors)
Concentration: Ecology and Evolutionary Biology

Grants and Fellowships

Animal Behavior Society. Unknown Caller: Identity-mediated Responses in a Communal and Cooperative Breeding Bird (2025). $2,000

Program in Latin American Studies, Princeton University. Unknown Caller: Identity-mediated Responses in a Communal and Cooperative Breeding Bird (2025). $3,000

APS Lewis and Clark Fund for Exploration and Field Research. Understanding the social structure of a cooperative and communal breeding bird in the Dominican Republic (2024). $5,000

Birds Caribbean David S. Lee Fund for the Conservation of Birds. Understanding the social and Spatial Organization of a Neotropical Bird (2024). $1,000

Princeton President's Fellow (2022). $6,000

Explorers Club Youth Activity Fund Rising Explorer Grant. Interbreeding of two subspecies in the Western Himalayas (2020). $2,000

Cornell Lab of Ornithology. Greenish Warbler Research with Trevor Price in
the Western Himalayas (2019). $1,500

Peer-Reviewed Publications

Kent, Q. D., LaPergola, J. B., Riehl, C. (*in review*). Spatial organization and breeding biology of communally nesting Palmchats (Dulus dominicus). *Journal of Field Ornithology*.

Kent, Q. D., LaPergola, J. B., Riehl, C. (*in prep*). Context-dependent use of mimetic and referential alarm calls by a cooperative songbird. *Target Journal: PNAS* .

Kent, Q. D., Villacorta, E., Riehl, C. (*in prep*). Individual, group, and regional signatures in the Palmchat (Dulus dominicus) distress call. *Target Journal: Royal Proceedings B*.

Lane, E., **Kent, Q. D.,** Agrawal, M., Pravosudov, V. V., Branch, C. L., (*in review*). Is it a signal? Achromatic plumage reflectance is associated with winter climatic harshness in a food-caching bird. *Behavioral Ecology and Sociobiology.*

Kent, Q. D., Edwards, M, Wu, T., Dhondt, A. A. (2020). Picky Palmchats (*Dulus Dominicus*): do they really prefer to nest in royal palms? *Journal of Caribbean* 33: 111-115.

Professional Presentations

Kent, **Q. D.,** LaPergola, J. B., Riehl, C (2026). Unknown Caller: Identity-mediated Responses in a Communal and Cooperative Breeding Bird. *International Society for Behavioral Ecology.* July 2026. (Oral Presentation).

Kent, **Q. D.,** LaPergola, J. B., Riehl, C (2024). Understanding the social and spatial organization of the Palmchat (Dulus dominicus). *BirdsCaribbean Meeting*. July 2024. (Oral Presentation).

Branch, C. L., Lane, E., **Kent, Q. D.** (2021). Variation in achromatic plumage patches along an elevation gradient. *Animal Behavior Society Virtual Meeting*. August 2021. (Oral Presentation).

Invited Talks

Kent, Q. D. (2025). Are You Going to Help? Understanding the role identity plays in coordinating cooperative behavior. *Peddie School, NJ*. April 2025.

Research Experience

Dissertation Chapter One, Spatial organization and breeding biology of communally nesting Palmchats (Dulus dominicus). September 2022 – March 2026

- Designed and executed a spatial point pattern analysis investigating drivers of nest site selection in a colonial-roosting bird species across a heterogeneous coastal landscape (~1,200 georeferenced data points)
- Applied inhomogeneous Ripley's K/L-functions and random labeling tests in R (spatstat) to disentangle first-order environmental heterogeneity from second-order biological interactions
- Detected and corrected for spatial autocorrelation (Moran's I) using spatial autoregressive (SAR) error models, improving inferential validity over standard GLMs
- Built binomial GLMs to identify morphological and spatial predictors of habitat selection, troubleshooting multicollinearity, separation, and missing-data issues
- Translated complex spatial statistics into accessible visualizations and written explanations for interdisciplinary audiences.

Dissertation Chapter Two, Context-dependent use of mimetic and referential alarm calls by a cooperative songbird. February 2026 – August 2026

- Designed and built a reproducible Python pipeline to test the acoustic fidelity of palmchat mimicry of Ridgway's Hawk vocalizations; manuscript submitted for peer review (August 2026)
- Collected and curated ~900 vocalizations from 40+ palmchat individuals and 10+ Ridgway's Hawk recording sessions across three field seasons in the Dominican Republic
- Implemented multi-method acoustic comparison framework: 20 predefined acoustic features (PAFs), MFCC + cosine distance, normalized DTW on log-STFT spectrograms
- Performed dimensionality reduction (UMAP on precomputed distance matrices), KNN-graph clustering with Louvain community detection and NMI scoring, and Random Forest classification (binary matched + 4-class) with leave-one-unit-out cross-validation
- Statistical inference: PERMANOVA with unit-level label permutation, Cohen's d, Holm-Bonferroni correction, linear mixed-effects models.

Independent Portfolio Project, Geospatial Analysis of California Herpetofauna Biodiversity and Wildfire Exposure. June 2026 - Present

- Built an end-to-end spatial analysis in ArcGIS Online assessing wildfire exposure across the ranges of 20 California endemic reptile and amphibian species, integrating CWHR species range data with CAL FIRE historic fire perimeters (1950–2025).
- Engineered a reproducible data pipeline in Python (Shapely) to merge, normalize, and dissolve 20 source GeoJSON datasets with inconsistent schemas into a single analysis-ready feature layer.
- Computed geodetic range and burned-area metrics using Arcade, quantifying the proportion of each species' range affected by fire over 5- and 10-year windows.
- Developed a species-richness surface via hexagonal tessellation and spatial joins to identify biodiversity hotspots, classifying them into restoration priorities (recently burned) and fire-regime-evaluation priorities (long-unburned).
- Framed results around fire-return-interval departure rather than fire avoidance, reflecting fire-adapted ecosystem dynamics.
- Published an interactive web map and dashboard communicating priority areas for conservation and fire management.

First Year Project, Leveraging Machine Learning Algorithms to Annotate Birdsong in East African Soundscapes. January 2023 – June 2023

- Deployed BirdNET deep neural network to automatically detect and classify bird vocalizations across 7 field sites spanning an urbanization gradient in Central Kenya, evaluating model performance in an understudied, real-world soundscape.
- Conducted statistical analysis in R (tidyverse, dplyr, data.table), including linear modeling of accuracy vs. confidence threshold and characterization of species-richness decay.

- Ran BirdNET via command line and systematically swept minimum confidence thresholds (0.1–0.8) to quantify classification accuracy against a ground-truth Kenyan species list.
- Owned the full pipeline end-to-end: field data collection, audio preprocessing, ML deployment, and statistical analysis.

Honor Thesis, Cornell University. May 2020- June 2022

- Formulated hypotheses to explain the change in avian diversity between sites in the northwestern Himalayas.
- Extracted data from the largest bird databases online including eBird, Birdlife International, and the Sheard et al. (2020) paper as well as from the world's largest climate database (WorldClim).
- Visualized the data in the form of graphs and figures using R.
- Performed linear models to determine which factors influenced the change in community composition.
- Interpreted the results of the analysis to write a scientific paper that will be submitted to an academic journal.

Cornell Lab of Ornithology, Lovette Lab and The Fuller Evolutionary Biology Program, February 2019-March 2020

- Measured plumage patterns of Mountain Chickadees using the software program Image J.
- Communicated with Post-doctoral advisor about tasks for the week.
- Incorporated lab techniques to extract DNA from samples collected at field sites.
- Discussed scientific papers with my advisor and other members of my lab during weekly meetings.

Tropical Field Ornithology, Punta Cana, Dominican Republic, Field Course. January 2020.

- Developed a hypothesis for the research project while leading a group of students during fieldwork.
- Collected and analyzed data on tree height, nest size, and nest site location of Palmchats using R.
- Captured and banded several Palmchats as well as recorded morphometric data while birds were in hand.
- Collaborated with Professor André Dhondt and students to publish a paper in the Caribbean Journal of Ornithology with me as the first author.

Price Research Lab, Himachal Pradesh, India, Field Assistant. May 2019-July 2019.

- Managed a group of 4 field assistants while collecting information on the Greenish Warbler.
- Recorded bird songs in the field and analyzed the recorded songs using the R program WarbleR.
- Orchestrated the placement and set up of mist nets to capture the target species.
- Extracted DNA from birds in the form of blood and feathers after birds were caught in nets.
- Documented important information about birds and the surrounding environment such as the location of nests and individual territories.

Braddock Bay Bird Observatory, Rochester, NY, Field Assistant, October 2019.

- Extracted birds from mist nets.

- Conducted and organized morphological measurements on birds removed from nets.
- Communicated with other members of the banding station to coordinate net checks.

Honors and Awards

Honorable Mention, NSF GRFP, 2024
Scholar, Richard M. Schulze Family Foundation, 2018-2022
Scholar, Vail Recreation District, 2018-2022
Scholar, Biology Scholars Program, 2018-2022
Award, American Ornithological Society Student Membership Award, 2020

Students Advised/Mentored

Paige Landry (2024-25). Chatty palmchats: The diverse vocal repertoire of Dulus dominicus.

Soloman Williams (2025-26). The Not So Golden Rule: Cooperation Within and Between Levels of a Multilevel Society as Observed in Palmchats (Dulus dominicus).

Ella Villacorta (2025-26). Fear in Many Voices: Evidence for Vocal Dialects in Palmchat Distress Calls.

Professional Memberships

- Member, International Society for Behavioral Ecology, 2025
- Member, Animal Behavior Society 2025
- Member, American Ornithological Society 2020
- Member, Birds Caribbean 2024

Technical/Field Skills

Wet Lab: I have performed DNA extractions, DNA assays, and gel electrophoresis.

Field: I have experience with point counts, line transects, nest searching, target and passive mist netting, bird banding, and audio playback experiments. I have used shotgun microphones and trail cameras. I have placed PIT tags on birds and set up RFID antennae.

Software: I have experience with R Studio (dyplyr, tidyverse, ggplot2), Python (librosa, scikit-learn, UMAP, HDBSCAN, statsmodels, pandas), Unix, SQL, ImageJ, BirdNet, Raven, Mr. Bayes, RAxML, and various population ecology programs (Populus and Maxima).

Version Control: Git/Github